

Andrew Shao

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering: Computer Science

Dec 2025

• Relevant Coursework:

Distributed Systems, Web Systems, Operating Systems, Applied Parallel Programming with GPUs, Machine Learning, Natural Language Processing, Applied Linear Algebra

WORK EXPERIENCE

Rockwell Automation | *Software Intern*

May 2025 - Aug 2025

- Built an AI agent using LangChain, FastAPI, and Apache Kafka to analyze 2M+ CI/CD logs and detect failures, reducing manual review time by 85% and streamlining debugging via automated GitLab issue creation with suggested code fixes.
- Navigated a large legacy codebase for Logix Designer to resolve 6 critical bugs in C++ and C# across concurrency, memory management, and legacy interface issues; utilized .NET profiling to optimize asynchronous resource management.
- Built an automated security tool in Python using GitLab CI/CD that integrates with the BlackDuck API to parse security vulnerability data and generate Jira tickets, streamlining the reporting workflow and improving triage efficiency by 20%.
- Developed C#/.NET 8 sample projects for the FactoryTalk Echo SDK enabling controller automation and co-simulation, while writing C++/C# unit and integration tests merged into a Jenkins CI/CD pipeline to improve release confidence.

Sigwise | *Software Intern*

May 2023 - Aug 2023

- Engineered a high-throughput, asynchronous gRPC and Protobuf telemetry pipeline in C++ to stream real-time application service metrics to a live observability dashboard, reducing monitoring collection latency from 200ms to 130ms.
- Designed and deployed Docker containers for backend alert services, utilizing Kubernetes for cluster management and scaling, cutting environment setup time from 2 hours to under 15 minutes and streamlining testing across the team.
- Developed a motion-detection Android security camera app using OpenCV, triggering video capture with intensity-based recording thresholds, and implemented a JavaMail/SMTP notification system to deliver snapshots and clips via email.

University of Michigan | *Student Computer Consultant*

Oct 2022 - Mar 2026

- Built custom log-parsing Bash scripts and Zabbix monitoring dashboards to track system health and network performance across 4 campus computing sites, reducing network troubleshooting time by 30% and preventing critical service disruptions.
- Built a Chrome browser extension in Javascript that restructured helpdesk ticket data into a condensed dashboard and automated repetitive workflow actions via custom macros, greatly reducing ticket triage time for 100+ support agents.

Hyland Software | *Software Intern (PEEKE Internship Program)*

Jun 2021 - Dec 2021

- Developed a document processing backend microservice using C++ and PostgreSQL, leveraging indexing to reduce document ingestion latency by 30% and enabling sub-second search retrieval, integrated into a CI/CD pipeline.
- Created a full-stack web application to host the next Hyland Hackathon using React.js, Django, and SQLite, designed to display real-time announcements, chat messages, and team data, allowing the hackathon to shift to a hybrid format.

PROJECTS

S-Watch: Movie Recommendation & Streaming Platform

Go, Gin-gonic, Python, ReactJS, GCP, MongoDB

- Built a self-trained movie recommendation engine using ALS matrix factorization and 100K MovieLens ratings as data, trained in Python and served from Go per-request with a 0.72 hit-rate at 10 vs. 0.62 popularity baseline.
- Developed a full-stack movie streaming platform with a Go/Gin backend and responsive React frontend, featuring JWT authentication and MongoDB Atlas for scalable storage; deployed on GCP Cloud Run and Vercel for high availability.

Distributed Key-Value Store (Paxos & Sharded KV Store)

Go, Paxos

- Built a resilient, horizontally scalable distributed key-value store in Go using Paxos consensus and a sharded architecture across 12 shards / 30+ nodes, ensuring linearizable consistency at 15,000 ops/sec with live shard migration.
- Designed an RPC communication layer with idempotent request handling and optimized routing, achieving at-most-once semantics with p99 latency under 10ms and zero data loss across 100+ simulated node-failure trials.

SKILLS

Languages: C++/C, Go, Python, Java, C#, TypeScript, JavaScript, SQL, Bash, Kotlin, HTML/CSS

Frameworks: React, Flask, Angular, Spring, .NET, OpenCV, TensorFlow, PyTorch, Pandas/NumPy

Cloud & Tools: Google Cloud Platform, AWS, Kafka, Docker, Kubernetes, Postgres, CI/CD, CUDA, Redis, Git, Linux